

Multiple VPC Networks - LAB

GSP211



Overview

Virtual Private Cloud (VPC) networks allow you to maintain isolated environments within a larger cloud structure, giving you granular control over data protection, network access, and application security.

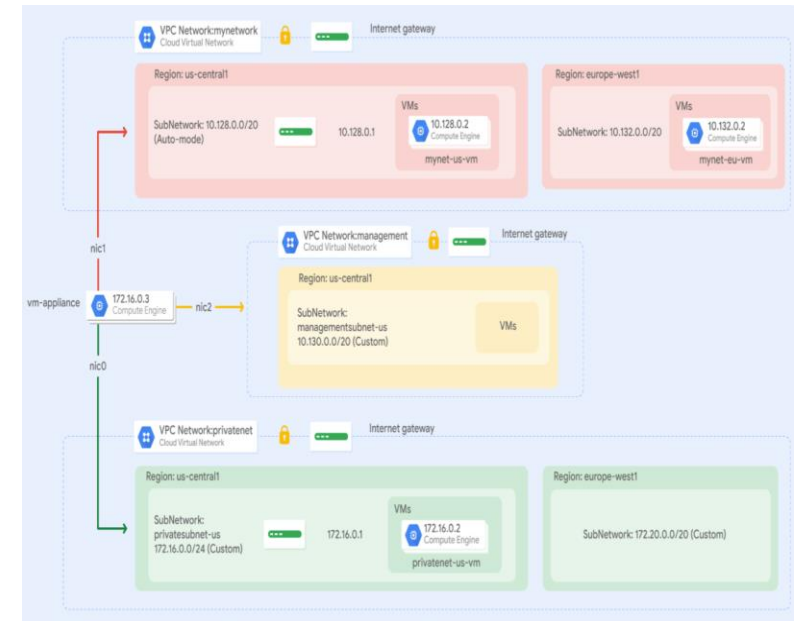
In this lab you create several VPC networks and VM instances, then test connectivity across networks. Specifically, you create two custom mode networks (**managementnet** and **privatenet**) with firewall rules and VM instances as shown in this network diagram:

The **mynetwork** network with its firewall rules and two VM instances (**mynet-vm-1** and **mynet-vm-2**) have already been created for you for this lab.

Objectives

In this lab, you will learn how to perform the following tasks:

- Create custom mode VPC networks with firewall rules
- Create VM instances using Compute Engine
- Explore the connectivity for VM instances across VPC networks
- Create a VM instance with multiple network interfaces



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Lab setup instructions and requirements

Protect your account and progress. Always use a private browser window and lab

End Lab

01:08:10

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Open Google Cloud console

Username

student-01-efbc7dd07c21f

Password

H0jjum9t8pUN

Project ID

qwiklabs-gcp-04-a50ab16c

Note: For full documentation of gcloud, in Google Cloud, refer to the gcloud CLI overview guide.

Task 1. Create custom mode VPC networks with firewall rules

Create two custom networks **managementnet** and **privatenet**, along with firewall rules to allow **SSH**, **ICMP**, and **RDP** ingress traffic.

Create the managementnet network

Create the **managementnet** network using the Cloud console.

1. In the Cloud console, navigate to **Navigation menu** (≡) > **VPC network** > **VPC networks**.

Google Cloud

bs-gcp-04-d5e0977455a7

Lab instructions and tasks

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Congratulations!

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	Shared VPC	☐	Name 	Subnets	MTU 	Mode	IPv6 ULA range	Gateways	Flow log configs 	Firewall rules	Global dynamic routing
	Serverless VPC access	☐	default	24	1460	Auto				4	Off
	Packet mirroring	☐	mynetwork	23	1460	Auto				3	Off

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Open Google Cloud console

Usernamestudent-01-efbc7dd07c21f

PasswordH0jjum9t8pUN

Project IDqwiklabs-gcp-04-a50ab16f

Multiple VPC Networks

Each Google Cloud project starts with the default network. In addition, the mynetwork network has been premade as part of your network diagram.

3. Click Create VPC Network.

4. Set the Name to managementnet.

5. For Subnet creation mode, click Custom.

6. Set the following values, leave all other values at their defaults:

Property	Value (type value or select option as specified)
Name	managementsubnet-1
Region	us-central1
IPv4 range	10.130.0.0/20

7. Click Done.

8. Click EQUIVALENT COMMAND LINE.

These commands illustrate that networks and subnets can be created using the Cloud Shell command line. You will create the privatenet network using these commands with similar parameters.

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VPC Network / VPC networks / Create VPC network

VPC networks

IP addresses

Internal ranges

Bring your own IP

Firewall

Routes

VPC network peering

Shared VPC

Serverless VPC access

Packet mirroring

VPC Flow Logs

←

Create a VPC network

Name *
managementnet
Lowercase letters, numbers, hyphens allowed

Description

Maximum transmission unit (MTU)
1460

☐ Configure network profile

Subnet creation mode
☒ Custom
☐ Automatic

Private IPv6 address settings
☐ Configure a ULA internal IPv6 range for this VPC Network
The ULA range is a /48 CIDR from which all private IPv6 subnet ranges will be taken. Google Cloud can allocate one automatically or you can allocate one manually. Allocation is permanent. You cannot deallocate or change the ULA range.

Subnets
Subnets let you create your own private cloud topology within Google Cloud. Click Automatic to create a subnet in each region, or click Custom to manually define the subnets. [Learn more](#)

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VPC networks

IP addresses

Internal ranges

Bring your own IP

Firewall

Routes

VPC network peering

Shared VPC

Serverless VPC access

Packet mirroring

VPC Flow Logs

← Create a VPC network

Name *
managementsubnet-1 ⓘ
Lowercase letters, numbers, hyphens allowed

Description

Region *
us-central1 (Iowa) ⓘ

IP stack type ⓘ

☒ IPv4 (single-stack)

☐ IPv4 and IPv6 (dual-stack)

☐ IPv6 (single-stack)

Primary IPv4 range

☐ Associate with an internal range
Use an internal range to specify the subnet's internal IP address range. The subnetwork can be associated with an entire internal range or only part of the range.

IPv4 range *
10.130.0.0/20 ⓘ
E.g. 10.0.0.0/24

Secondary IPv4 ranges ⓘ

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End Lab01:03:59

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Open Google Cloud console

Username

student-01-efbc7dd07c214

Password

H0jjum9t8pUN

Project ID

qwiklabs-gcp-04-a50ab16:

IPv4 range10.130.0.0/20

7. Click Done.

8. Click EQUIVALENT COMMAND LINE.

These commands illustrate that networks and subnets can be created using the Cloud Shell command line. You will create the privatenet network using these commands with similar parameters.

9. Click Close.

10. Click Create.

Test Completed Task

Click Check my progress to verify your performed task. If you have successfully created a managementnet network, you will see an assessment score.

Create the managementnet network

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VPC networks

+ Create VPC network

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📄 Networks in current project

📄 Subnets in current project

📘 SMTP port 25 disallowed in this project. [Learn more](#)

VPC networks

☰ Manage flow logs

☰ Filter

Enter property name or value

?

☰

☐	Name ↑	Subnets	MTU ?	Mode	IPv6 ULA range	Gateways	Flow log configs ?	Firewall rules	Global dynamic routing	Network profil
☐	default	24	1460	Auto				4	Off	
☐	managementnet	1	1460	Custom				0	Off	
☐	mynetwork	23	1460	Auto				3	Off	

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Successfully created network "managementnet".

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Open Google Cloud console

Username

student-01-efbc7dd07c214

Password

H0jjum9t8pUN

Project ID

qwiklabs-gcp-04-a50ab16

Create the privatenet network

Create the **privatenet** network using the Cloud Shell command line.

1. Run the following command to create the **privatenet** network:

gcloud compute networks create privatenet --subnet-mode=custom

2. Run the following command to create the **privatesubnet-1** subnet:

gcloud compute networks subnets create privatesubnet-1 --network=privatenet --region=us-central1 --range=172.16.0.0/24

3. Run the following command to create the **privatesubnet-2** subnet:

gcloud compute networks subnets create privatesubnet-2 --

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End Lab

01:02:24

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

Open Google Cloud console

Username

student-01-efbc7dd07c214

Password

H0jjum9t8pUN

Project ID

qwiklabs-gcp-04-a50ab16c

Create the **privatenet** network using the Cloud Shell command line.

1. Run the following command to create the **privatenet** network:

gcloud compute networks create privatenet --subnet-mode=custom

2. Run the following command to create the **privatesubnet-1** subnet:

gcloud compute networks subnets create privatesubnet-1 --network=privatenet --region=us-central1 --range=172.16.0.0/24

3. Run the following command to create the **privatesubnet-2** subnet:

gcloud compute networks subnets create privatesubnet-2 --network=privatenet --region=europe-west1 --range=172.20.0.0/20

Test Completed Task

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Congratulations!

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📘 Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#)

Don't show again Dismiss

```
project = qwiklabs-gcp-04-a50ab163c625

Your active configuration is: [cloudshell-9408]
student_01_efbc7dd07c21@cloudshell:~ (qwiklabs-gcp-04-a50ab163c625)$ gcloud compute networks create privatenet --subnet-mode=custom
Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-04-a50ab163c625/global/networks/privatenet].
NAME: privatenet
SUBNET_MODE: CUSTOM
BGP_ROUTING_MODE: REGIONAL
IPV4_RANGE:
GATEWAY_IPV4:
INTERNAL_IPV6_RANGE:

Instances on this network will not be reachable until firewall rules
are created. As an example, you can allow all internal traffic between
instances as well as SSH, RDP, and ICMP by running:

$ gcloud compute firewall-rules create <FIREWALL_NAME> --network privatenet --allow tcp,udp,icmp --source-ranges <IP_RANGE>
$ gcloud compute firewall-rules create <FIREWALL_NAME> --network privatenet --allow tcp:22,tcp:3389,icmp

student_01_efbc7dd07c21@cloudshell:~ (qwiklabs-gcp-04-a50ab163c625)$ gcloud compute networks subnets create privatesubnet-1 --network=privatenet --region=us-central1 --range=172.16.0.0/24
Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-04-a50ab163c625/regions/us-central1/subnetworks/privatesubnet-1].
NAME: privatesubnet-1
REGION: us-central1
NETWORK: privatenet
RANGE: 172.16.0.0/24
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:
student_01_efbc7dd07c21@cloudshell:~ (qwiklabs-gcp-04-a50ab163c625)$ gcloud compute networks subnets create privatesubnet-2 --network=privatenet --region=europe-west1 --range=172.20.0.0/20
Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-04-a50ab163c625/regions/europe-west1/subnetworks/privatesubnet-2].
NAME: privatesubnet-2
REGION: europe-west1
NETWORK: privatenet
RANGE: 172.20.0.0/20
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:
student_01_efbc7dd07c21@cloudshell:~ (qwiklabs-gcp-04-a50ab163c625)$
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End Lab01:00:27

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Open Google Cloud console

Username

student-01-efbc7dd07c214

Password

H0jjum9t8pUN

Project ID

qwiklabs-gcp-04-a50ab16

Create the privatenet network

Check my progress

Assessment Completed!

4. Run the following command to list the available VPC networks:

gcloud compute networks list

The output should look like this:

NAME: default

SUBNET_MODE: AUTO

BGP_ROUTING_MODE: REGIONAL

IPV4_RANGE:

GATEWAY_IPV4:

NAME: managementnet

Check complete. Points earned: 30. Message: Assessment Completed!

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Objectives

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Congratulations!

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```
EXTERNAL_IPV6_PREFIX:
student_01_efbc7dd07c21@cloudshell:~ (qwiklabs-gcp-04-a50ab163c625)$ gcloud compute networks subnets create privatesubnet-2 --network=privatenet --region=europe-west1 --range=172.20.0.0/20
Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-04-a50ab163c625/regions/europe-west1/subnetworks/privatesubnet-2].
NAME: privatesubnet-2
REGION: europe-west1
NETWORK: privatenet
RANGE: 172.20.0.0/20
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:
student_01_efbc7dd07c21@cloudshell:~ (qwiklabs-gcp-04-a50ab163c625)$ gcloud compute networks list
NAME: default
SUBNET_MODE: AUTO
BGP_ROUTING_MODE: REGIONAL
IPV4_RANGE:
GATEWAY_IPV4:
INTERNAL_IPV6_RANGE:

NAME: managementnet
SUBNET_MODE: CUSTOM
BGP_ROUTING_MODE: REGIONAL
IPV4_RANGE:
GATEWAY_IPV4:
INTERNAL_IPV6_RANGE:

NAME: mynetwork
SUBNET_MODE: AUTO
BGP_ROUTING_MODE: REGIONAL
IPV4_RANGE:
GATEWAY_IPV4:
INTERNAL_IPV6_RANGE:

NAME: privatenet
SUBNET_MODE: CUSTOM
BGP_ROUTING_MODE: REGIONAL
IPV4_RANGE:
GATEWAY_IPV4:
INTERNAL_IPV6_RANGE:
student_01_efbc7dd07c21@cloudshell:~ (qwiklabs-gcp-04-a50ab163c625)$
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Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Username

student-01-efbc7dd07c214

Password

H0jjum9t8pUN

Project ID

qwiklabs-gcp-04-a50ab16:

Note: default and mynetwork are auto mode networks, whereas, managementnet and privatenet are custom mode networks. Auto mode networks create subnets in each region automatically, while custom mode networks start with no subnets, giving you full control over subnet creation

5. Run the following command to list the available VPC subnets (sorted by VPC network):

gcloud compute networks subnets list --sort-by=NETWORK

The output should look like this:

NAME: default
REGION: us-central1
NETWORK: default
RANGE: 10.128.0.0/20
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:

Objectives

Setup and requirements

Task 1. Create custom mode VPC networks with firewall rules

Task 2. Create VM instances

Task 3. Explore the connectivity between VM instances

Task 4. Create a VM instance with multiple network interfaces

Congratulations!

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📘 Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#)

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```
GATEWAY_IPV4:
INTERNAL_IPV6_RANGE:
student_01_efbc7dd07c21@cloudshell:~ (qwiklabs-gcp-04-a50ab163c625)$ gcloud compute networks subnets list --sort-by=NETWORK
NAME: default
REGION: us-central1
NETWORK: default
RANGE: 10.128.0.0/20
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:
UTILIZATION_DETAILS:

NAME: default
REGION: europe-west1
NETWORK: default
RANGE: 10.132.0.0/20
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:
UTILIZATION_DETAILS:

NAME: default
REGION: us-west1
NETWORK: default
RANGE: 10.138.0.0/20
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:
UTILIZATION_DETAILS:

NAME: default
REGION: asia-east1
NETWORK: default
RANGE: 10.140.0.0/20
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
```

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VPC Network / VPC networks

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📄 IP addresses

📄 Internal ranges

🌐 Bring your own IP

🔒 Firewall

🔗 Routes

🔗 VPC network peering

🔗 Shared VPC

🔗 Serverless VPC access

🔗 Packet mirroring

☰ VPC Flow Logs

<|

📄 Networks in current project

📄 Subnets in current project

📄 VPC networks

☰ Manage flow logs

☰ Filter

Enter property name or value

?

☰

☐	Name ↑	Subnets	MTU ?	Mode	IPv6 ULA range	Gateways	Flow log configs ?	Firewall rules	Global dynamic routing	Network profil
☐	default	24	1460	Auto				4	Off	
☐	managementnet	1	1460	Custom				0	Off	
☐	mynetwork	23	1460	Auto				3	Off	
☐	privatenet	2	1460	Custom				0	Off	

📄 SMTP port 25 disallowed in this project. [Learn more](#)

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STACK_TYPE: IPV4_ONLY

IPV6_ACCESS_TYPE:

INTERNAL_IPV6_PREFIX:

EXTERNAL_IPV6_PREFIX:

UTILIZATION_DETAILS:

student_01_eFbc7dd07c21@cloudshell:~ (qwiklabs-gcp-04-a50ab163c625) \$

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53360

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Multiple VPC Networks

Lab setup instructions and requirements

End Lab00:57:52

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Usernamestudent-01-efbc7dd07c214

PasswordH0jjum9t8pUN

Project IDqwiklabs-gcp-04-a50ab16

Create firewall rules to allow SSH, ICMP, and RDP ingress traffic to VM instances on the managementnet network.

1. In the Cloud console, navigate to Navigation menu (≡) > VPC network > Firewall.

2. Click + Create Firewall Rule.

3. Set the following values, leave all other values at their defaults:

Property	Value (type value or select option as specified)
Name	managementnet-allow-icmp-ssh-rdp
Network	managementnet
Targets	All instances in the network
Source filter	IPv4 Ranges
Source IPv4 ranges	0.0.0.0/0

Objectives

Setup and requirements

Task 1. Create custom mode VPC networks with firewall rules

Task 2. Create VM instances

Task 3. Explore the connectivity between VM instances

Task 4. Create a VM instance with multiple network interfaces

Congratulations!

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VPC networks

📄 Networks in current project

📄 Subnets in current project

📄 SMTP port 25 disallowed in this project. [Learn more](#)

VPC networks

☰ Manage flow logs

☰ Filter Enter property name or value

☐	Name ↑	Subnets	MTU ?	Mode	IPv6 ULA range	Gateways	Flow log configs ?	Firewall rules	Global dynamic routing	Network profil
☐	default	24	1460	Auto				4	Off	
☐	managementnet	1	1460	Custom				0	Off	
☐	mynetwork	23	1460	Auto				3	Off	
☐	privatenet	2	1460	Custom				0	Off	

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STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:
UTILIZATION_DETAILS:
<https://console.cloud.google.com/networking/firewalls/list?cloudshell=true&project=qwikla...>

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Create a firewall rule – Network

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Cloud Armor policies

Adaptive Protection

Cloud Armor Service Tier

Cloud IDS

IDS Dashboard

IDS Endpoints

IDS Threats

Cloud NGFW

Dashboard

Firewall policies

Threats

Firewall endpoints

Secure Access Connect

Realms

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Create a firewall rule

🔍🔒

Firewall rules control incoming and outgoing traffic to an instance. By default, all incoming traffic to your network is blocked. [Learn more](#)

Name *

managementnet-allow-icmp-ssh-rdp

Lowercase letters, numbers, hyphens allowed

Description

Logs

Turning on firewall logs can generate a large number of logs which can increase costs in Logging. [Learn more](#)

☐ On

☒ Off

Network *

managementnet

Priority *

1000

Compare

Priority can be 0 - 65535

Direction of traffic

☒ Ingress

☐ Egress

Action on match

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Preview

Create a firewall rule

Protocols and ports

Allow all

Specified protocols and ports

TCP

Ports

22,3389

E.g. 20, 50-60

UDP

Ports

E.g. all

SCTP

Ports

E.g. 20, 50-60

Other

Protocols *

icmp

Separate multiple protocols by commas, e.g. ah, icmp

Disable rule

Create

Cancel

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Create a firewall rule – Network

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Allow all

Specified protocols and p

☒ TCP

Ports

22,3389

E.g. 20, 50-60

☐ UDP

Ports

E.g. all

☐ SCTP

Ports

E.g. 20, 50-60

☒ Other

Protocols *

icmp

Separate multiple protocols t

Disable rule

Create

Cancel

Equivalent command line

gcloud command line

This is the gcloud command line with the parameters you have selected. [gcloud reference](#)

```
$ gcloud compute --project=qwiklabs-gcp-04-a50ab163c625 firewall-rules create managementnet-allow-icmp-ssh-rdp --direction=INGRESS --priority=1000 --network=managementnet --action=ALLOW --rules=tcp:22,tcp:3389 --source-ranges=0.0.0.0/0
```

☒ Line wrapping

Copy all to clipboard

Run in Cloud Shell

Close

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🏠> Develop Your Google Cloud Network > Multiple VPC Networks

Contents

Multiple VPC Networks

Lab setup instructions and requirements

End Lab00:51:53

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Username

student-01-efbc7dd07c214

Password

H0jjum9t8pUN

Project ID

qwiklabs-gcp-04-a50ab16

Create the firewall rules for privatenet

Create the firewall rules for **privatenet** network using the Cloud Shell command line.

1. In Cloud Shell, run the following command to create the **privatenet-allow-icmp-ssh-rdp** firewall rule:

```
gcloud compute firewall-rules create privatenet-allow-icmp-ssh-rdp --direction=INGRESS --priority=1000 --network=privatenet --action=ALLOW --rules=icmp,tcp:22,tcp:3389 --source-ranges=0.0.0.0/0
```

The output should look like this:

```
Creating firewall...done.
NAME: privatenet-allow-icmp-ssh-rdp
NETWORK: privatenet
DIRECTION: INGRESS
```

Objectives

Setup and requirements

Task 1. Create custom mode VPC networks with firewall rules

Task 2. Create VM instances

Task 3. Explore the connectivity between VM instances

Task 4. Create a VM instance with multiple network interfaces

Congratulations!

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📘 Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#)

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```
NAME: privatesubnet-1
REGION: us-central1
NETWORK: privatenet
RANGE: 172.16.0.0/24
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:
UTILIZATION_DETAILS:

NAME: privatesubnet-2
REGION: europe-west1
NETWORK: privatenet
RANGE: 172.20.0.0/20
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:
UTILIZATION_DETAILS:
student_01_efbc7dd07c21@cloudshell:~ (qwiklabs-gcp-04-a50ab163c625)$ gcloud compute firewall-rules create privatenet-allow-icmp-ssh-rdp --direction=INGRESS --priority=1000 --network=privatenet --action=ALLOW --rules=icmp,tcp:22,tcp:3389 --source-ranges=0.0.0.0/0
Creating firewall...working..Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-04-a50ab163c625/global/firewalls/privatenet-allow-icmp-ssh-rdp].
Creating firewall...done.
NAME: privatenet-allow-icmp-ssh-rdp
NETWORK: privatenet
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: icmp,tcp:22,tcp:3389
DENY:
DISABLED: False
student_01_efbc7dd07c21@cloudshell:~ (qwiklabs-gcp-04-a50ab163c625)$
student_01_efbc7dd07c21@cloudshell:~ (qwiklabs-gcp-04-a50ab163c625)$
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student_01_efbc7dd07c21@cloudshell:~ (qwiklabs-gcp-04-a50ab163c625)$
student_01_efbc7dd07c21@cloudshell:~ (qwiklabs-gcp-04-a50ab163c625)$
```

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```
student_01_efbc7dd07c21@cloudshell:~ (qwiklabs-gcp-04-a50ab163c625)$
student_01_efbc7dd07c21@cloudshell:~ (qwiklabs-gcp-04-a50ab163c625)$ gcloud compute firewall-rules list --sort-by=NETWORK
NAME: default-allow-icmp
NETWORK: default
DIRECTION: INGRESS
PRIORITY: 65534
ALLOW: icmp
DENY:
DISABLED: False

NAME: default-allow-internal
NETWORK: default
DIRECTION: INGRESS
PRIORITY: 65534
ALLOW: tcp:0-65535,udp:0-65535,icmp
DENY:
DISABLED: False

NAME: default-allow-rdp
NETWORK: default
DIRECTION: INGRESS
PRIORITY: 65534
ALLOW: tcp:3389
DENY:
DISABLED: False

NAME: default-allow-ssh
NETWORK: default
DIRECTION: INGRESS
PRIORITY: 65534
ALLOW: tcp:22
DENY:
DISABLED: False

NAME: managementnet-allow-icmp-ssh-rdp
NETWORK: managementnet
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: tcp:22,tcp:3389,icmp
DENY:
DISABLED: False
```

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📘 Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#)

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```
PRIORITY: 1000
ALLOW: tcp:22,tcp:3389,icmp
DENY:
DISABLED: False

NAME: mynetwork-allow-icmp
NETWORK: mynetwork
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: icmp
DENY:
DISABLED: False

NAME: mynetwork-allow-rdp
NETWORK: mynetwork
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: tcp:3389
DENY:
DISABLED: False

NAME: mynetwork-allow-ssh
NETWORK: mynetwork
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: tcp:22
DENY:
DISABLED: False

NAME: privatenet-allow-icmp-ssh-rdp
NETWORK: privatenet
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: icmp,tcp:22,tcp:3389
DENY:
DISABLED: False

To show all fields of the firewall, please show in JSON format: --format=json
To show all fields in table format, please see the examples in --help.

student_01_efbc7dd07c21@cloudshell:~ (qwiklabs-gcp-04-a50ab163c625)$
```

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Firewall policies – Network Security

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Cloud Armor

- DDoS Dashboard
- Cloud Armor policies
- Adaptive Protection
- Cloud Armor Service Tier

Cloud IDS

- IDS Dashboard
- IDS Endpoints
- IDS Threats

Cloud NGFW

- Dashboard
- Firewall policies
- Threats
- Firewall endpoints

Secure Access Connect

- Realms Preview

Firewall policies

Create firewall policy | Create firewall rule

Filter: Enter property name or value

	Name	Type	Targets	Filters	Protocols / ports	Action	Priority	Network	Logs	Hit count	Last	
☐	default-allow-icmp	Ingress	Apply to all	IP ranges:	icmp	Allow	65534	default	Off	–	–	▼
☐	default-allow-internal	Ingress	Apply to all	IP ranges:	tcp:0-65535 udp:0-65535 icmp	Allow	65534	default	Off	–	–	▼
☐	default-allow-rdp	Ingress	Apply to all	IP ranges:	tcp:3389	Allow	65534	default	Off	–	–	▼
☐	default-allow-ssh	Ingress	Apply to all	IP ranges:	tcp:22	Allow	65534	default	Off	–	–	▼
☐	managementnet-allow-icmp-ssh-rdp	Ingress	Apply to all	IP ranges:	tcp:22, 3389 icmp	Allow	1000	managementnet	Off	–	–	▼
☐	mynetwork-allow-icmp	Ingress	Apply to all	IP ranges:	icmp	Allow	1000	mynetwork	Off	–	–	▼
☐	mynetwork-allow-rdp	Ingress	Apply to all	IP ranges:	tcp:3389	Allow	1000	mynetwork	Off	–	–	▼
☐	mynetwork-allow-ssh	Ingress	Apply to all	IP ranges:	tcp:22	Allow	1000	mynetwork	Off	–	–	▼
☐	privatenet-allow-icmp-ssh-rdp	Ingress	Apply to all	IP ranges:	tcp:22, 3389 icmp	Allow	1000	privatenet	Off	–	–	▼

Network firewall policies

Firewall policies let you group several firewall rules so that you can update them all at once, effectively controlled by Identity and Access Management (IAM) roles.

Learn more

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What do you want to learn today?

Multiple VPC Networks

Task 2. Create VM instances

Create two VM instances:

managementnet-vm-1 in managementsubnet-1

privatenet-vm-1 in privatesubnet-1

Create the managementnet-vm-1 instance

Create the managementnet-vm-1 instance using the Cloud console.

1. In the Cloud console, navigate to Navigation menu > Compute Engine > VM instances.

The mynet-vm-2 and mynet-vm-1 has been created for you, as part of your network diagram.

2. Click Create Instance.

3. In the Machine configuration:

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Multiple VPC Networks

Lab setup instructions and requirements

End Lab00:49:34

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Username

student-01-efbc7dd07c21d

Password

H0jjum9t8pUN

Project ID

qwiklabs-gcp-04-a50ab16c

Objectives

Setup and requirements

Task 1. Create custom mode VPC networks with firewall rules

Task 2. Create VM instances

Task 3. Explore the connectivity between VM instances

Task 4. Create a VM instance with multiple network interfaces

Congratulations!

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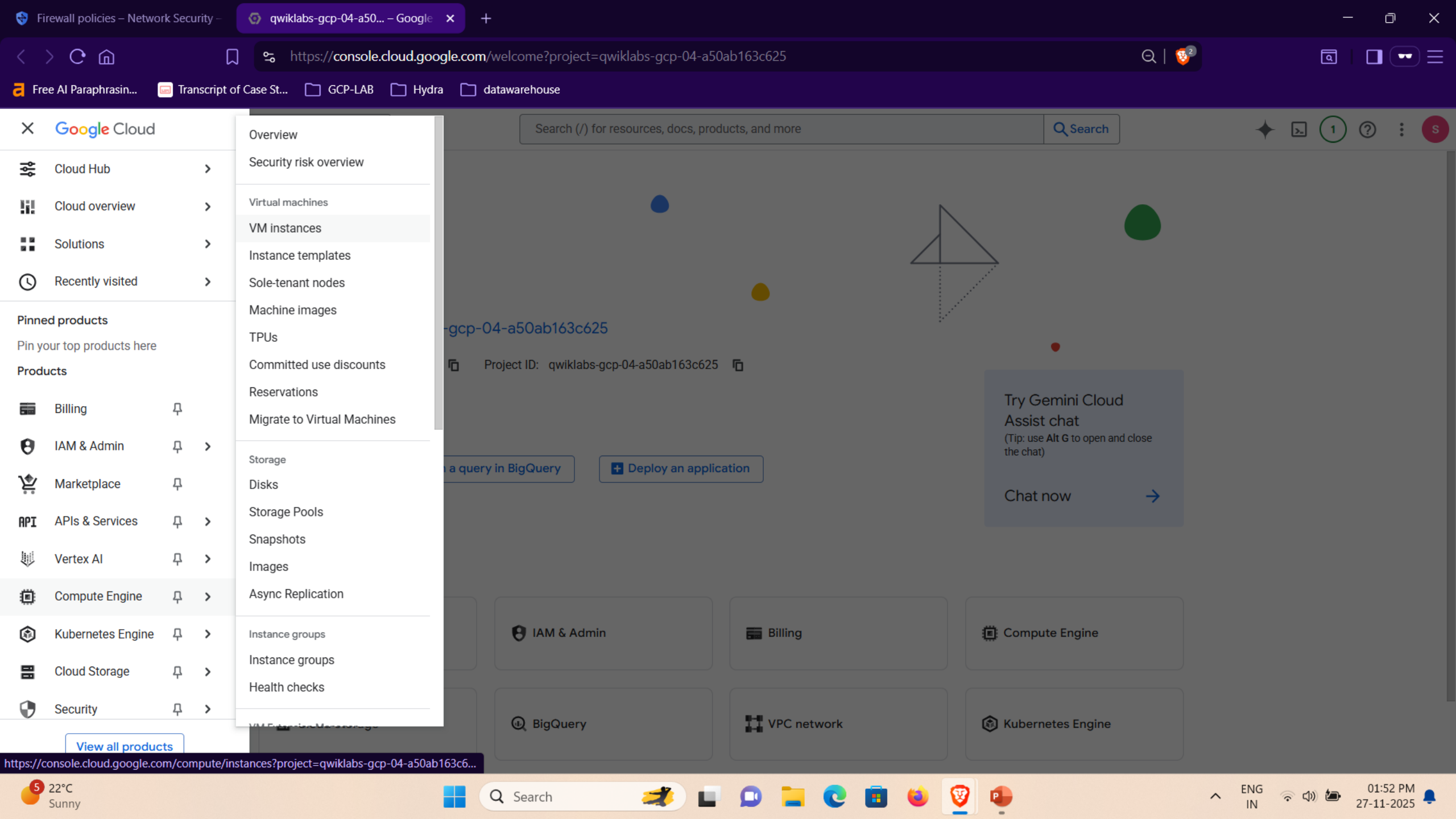
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Filter

Enter property name or value

Status	Name	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☒	mynet-vm-1	us-central1-c			10.128.0.2 (nic0)	34.132.85.179 (nic0)	SSH
☒	mynet-vm-2	eu-west-1-b			10.132.0.2 (nic0)	34.78.241.167 (nic0)	SSH

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Lab setup instructions and requirements

End Lab00:48:17

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Open Google Cloud console

Username

student-01-efbc7dd07c214

Password

H0jjum9t8pUN

Project ID

qwiklabs-gcp-04-a50ab16

instances.

The mynet-vm-2 and mynet-vm-1 has been created for you, as part of your network diagram.

2. Click Create Instance.

3. In the Machine configuration:

Set the following values, leave all other values at their defaults:

Property	Value (type value or select option as specified)
Name	managementnet-vm-1
Region	us-central1
Zone	us-central1-c
Series	E2
Machine Type	e2-micro

4. Click Networking.

Objectives

Setup and requirements

Task 1. Create custom mode VPC networks with firewall rules

Task 2. Create VM instances

Task 3. Explore the connectivity between VM instances

Task 4. Create a VM instance with multiple network interfaces

Congratulations!

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← Create an instance

🌟 Create VM from...

🔗 Equivalent code

• Machine configuration
e2-micro, us-central1-c

• OS and storage
Debian GNU/Linux 12 (bookworm)

• Data protection
Snapshot schedules

• Networking
1 network interface

• Observability
Install Ops Agent

• Security

• Advanced

N4

Flexible & cost-optimized

2 - 80

4 - 640 GB

Intel Emerald

N4D

Flexible & cost-optimized

2 - 96

4 - 768 GB

AMD Turin

C3

Consistently high performance

4 - 192

8 - 1,536 GB

Intel Sapphire

C3D

Consistently high performance

4 - 360

8 - 2,880 GB

AMD Genoa

E2

Low cost, day-to-day computing

0.25 - 32

1 - 128 GB

Intel Broadwe

N2

Balanced price & performance

2 - 128

2 - 864 GB

Intel Cascade

N2D

Balanced price & performance

2 - 224

2 - 896 GB

AMD Milan

T2A

Scale-out workloads

1 - 48

4 - 192 GB

Ampere Altra

T2D

Scale-out workloads

1 - 60

4 - 240 GB

AMD Milan

N1

Balanced price & performance

0.25 - 96

0.6 - 624 GB

Intel Haswell

Machine type

Choose a machine type with preset amounts of vCPUs and memory that suit most workloads. Or, you can create a custom machine for your workload's particular needs. [Learn more](#)

Preset

Custom

e2-micro (2 vCPU, 1 core, 1 GB memory)

vCPU

0.25-2 vCPU (1 shared core)

Memory

1 GB

📄

Advanced configurations

Monthly estimate

\$7.11

That's about \$0.01 hourly

Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
2 vCPU + 1 GB memory	\$6.11
10 GB balanced persistent disk	\$1.00
Logging	Cost varies
Monitoring	Cost varies
Snapshot schedule	Cost varies
Total	\$7.11

[Compute Engine pricing](#)

[Cloud Operations pricing](#)

Less

Create

Cancel

🔗 Equivalent code

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← Create an instance

🔧 Create VM from...

Equivalent code

Machine configuration

e2-micro, us-central1-c

OS and storage

Debian GNU/Linux 12 (bookworm)

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1 network interface, managementsubnet-1 (10.130.0.0/20)

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Security

Advanced

IP forwarding ?

☐ Enable

Network performance configuration

Network bandwidth ?

☐ Enable per VM Tier_1 networking performance

Maximum outbound network bandwidth: 1 Gbps

VM to Public IP: 1 Gbps

Network interfaces ?

Network interface is permanent

▼ nic0 managementnet managementsubnet-1 IPv4 (10.130.0.0/20)

🗑

Add a network interface

Dynamic Network Interfaces ?

Add a Dynamic Network Interface

Monthly estimate

\$7.11

That's about \$0.01 hourly

Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
2 vCPU + 1 GB memory	\$6.11
10 GB balanced persistent disk	\$1.00
Logging	Cost varies
Monitoring	Cost varies
Snapshot schedule	Cost varies
Total	\$7.11

[Compute Engine pricing](#)

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Equivalent code

Command line

REST

Terraform

```
1 gcloud compute instances create
  managementnet-vm-1 \
2   --project=qwiklabs-gcp-04-a50ab163c625 \
3   --zone=us-central1-c \
4   --machine-type=e2-micro \
5   --network-interface=network-tier=PREMIUM,
  stack-type=IPv4_ONLY,subnet=managementsubnet-1 \
6   --metadata=enable-osconfig=TRUE,
  enable-oslogin=true \
7   --maintenance-policy=MIGRATE \
8   --provisioning-model=STANDARD \
9
10  --service-account=500250487414-compute@developer.
  gserviceaccount.com \
11  --scopes=https://www.googleapis.com/auth/
  devstorage.read_only,https://www.googleapis.com/
  auth/logging.write,https://www.googleapis.com/
  auth/monitoring.write,https://www.googleapis.com/
  auth/service.management.readonly,https://www.
  googleapis.com/auth/servicecontrol,https://www.
  googleapis.com/auth/trace.append \
12  --create-disk=auto-delete=yes,boot=yes,
  device-name=managementnet-vm-1,image=projects/
  debian-cloud/global/images/
  debian-12-bookworm-v20251111,mode=rw,size=10,
  type=pd-balanced \
13  --no-shielded-secure-boot \
14  --shielded-vtpm \
15  --shielded-integrity-monitoring \
16
17  --labels=goog-ops-agent-policy=v2-x86-template-1-4
  -0,goog-ec-src=vm_add-gcloud \
18  --reservation-affinity=any \
  && \
  printf 'agentsRule:\n  packageState: installed\n
  version: latest\ninstanceFilter:\n
  inclusionLabels:\n  - labels:\n
```

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Run in Cloud Shell

View gcloud reference

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Filter

Enter property name or value

Status	Name	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☒	managementnet-vm-1	us-central1-c			10.130.0.2 (nic0)	34.28.248.151 (nic0)	SSH
☒	mynet-vm-1	us-central1-c			10.128.0.2 (nic0)	34.132.85.179 (nic0)	SSH
☒	mynet-vm-2	europa-west1-b			10.132.0.2 (nic0)	34.78.241.167 (nic0)	SSH

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Lab setup instructions and requirements

End Lab 00:43:34

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Username student-01-efbc7dd07c21d

Password H0jjum9t8pUN

Project ID qwiklabs-gcp-04-a50ab16

Assessment Completed!

Create the privatenet-vm-1 instance

Create the **privatenet-vm-1** instance using the Cloud Shell command line.

1. In Cloud Shell, run the following command to create the **privatenet-vm-1** instance:

gcloud compute instances create privatenet-vm-1 --zone=us-central1-c --machine-type=e2-micro --subnet=privatesubnet-1

The output should look like this:

Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-04-97] NAME: privatenet-vm-1 ZONE: us-central1-c

Objectives

Setup and requirements

Task 1. Create custom mode VPC networks with firewall rules

Task 2. Create VM instances

Task 3. Explore the connectivity between VM instances

Task 4. Create a VM instance with multiple network interfaces

Congratulations!

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Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#)

Don't show again

Dismiss

```
ALLOW: icmp
DENY:
DISABLED: False

NAME: mynetwork-allow-rdp
NETWORK: mynetwork
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: tcp:3389
DENY:
DISABLED: False

NAME: mynetwork-allow-ssh
NETWORK: mynetwork
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: tcp:22
DENY:
DISABLED: False

NAME: privatenet-allow-icmp-ssh-rdp
NETWORK: privatenet
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: icmp,tcp:22,tcp:3389
DENY:
DISABLED: False

To show all fields of the firewall, please show in JSON format: --format=json
To show all fields in table format, please see the examples in --help.

student_01_efbc7dd07c21@cloudshell:~ (qwiklabs-gcp-04-a50ab163c625)$ gcloud compute instances create privatenet-vm-1 --zone=us-central1-c --machine-type=e2-micro --subnet=privatesubnet-1
Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-04-a50ab163c625/zones/us-central1-c/instances/privatenet-vm-1].
NAME: privatenet-vm-1
ZONE: us-central1-c
MACHINE_TYPE: e2-micro
PREEMPTIBLE:
INTERNAL_IP: 172.16.0.2
EXTERNAL_IP: 34.59.31.233
STATUS: RUNNING
student_01_efbc7dd07c21@cloudshell:~ (qwiklabs-gcp-04-a50ab163c625)$
```

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Multiple VPC Networks

Lab setup instructions and requirements

End Lab

00:42:39

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Username

student-01-efbc7dd07c214

Password

H0jjum9t8pUN

Project ID

qwiklabs-gcp-04-a50ab16

Create the privatenet-vm-1 instance

Check my progress

Assessment Completed!

2. Run the following command to list all the VM instances (sorted by zone):

gcloud compute instances list --sort-by=ZONE

The output should look like this:

NAME: mynet-vm-2

ZONE: europe-west1-b

MACHINE_TYPE: e2-micro

PREEMPTIBLE:

INTERNAL_IP: 10.164.0.2

EXTERNAL_IP: 34.147.23.235

STATUS: RUNNING

Objectives

Setup and requirements

Task 1. Create custom mode VPC networks with firewall rules

Task 2. Create VM instances

Task 3. Explore the connectivity between VM instances

Task 4. Create a VM instance with multiple network interfaces

Congratulations!

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Don't show again

Dismiss

```
Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-04-a50ab163c625/zones/us-central1-c/instances/privatenet-vm-1].
NAME: privatenet-vm-1
ZONE: us-central1-c
MACHINE_TYPE: e2-micro
PREEMPTIBLE:
INTERNAL_IP: 172.16.0.2
EXTERNAL_IP: 34.59.31.233
STATUS: RUNNING
student_01_efbc7dd07c21@cloudshell:~ (qwiklabs-gcp-04-a50ab163c625)$ gcloud compute instances list --sort-by=ZONE
NAME: mynet-vm-2
ZONE: europe-west1-b
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.132.0.2
EXTERNAL_IP: 34.78.241.167
STATUS: RUNNING

NAME: managementnet-vm-1
ZONE: us-central1-c
MACHINE_TYPE: e2-micro
PREEMPTIBLE:
INTERNAL_IP: 10.130.0.2
EXTERNAL_IP: 34.28.248.151
STATUS: RUNNING

NAME: mynet-vm-1
ZONE: us-central1-c
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.128.0.2
EXTERNAL_IP: 34.132.85.179
STATUS: RUNNING

NAME: privatenet-vm-1
ZONE: us-central1-c
MACHINE_TYPE: e2-micro
PREEMPTIBLE:
INTERNAL_IP: 172.16.0.2
EXTERNAL_IP: 34.59.31.233
STATUS: RUNNING
student_01_efbc7dd07c21@cloudshell:~ (qwiklabs-gcp-04-a50ab163c625)$
```

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VM instances – Compute Engine

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https://console.cloud.google.com/compute/instances?onCreate=true&project=qwiklabs-gcp-04-a50ab163c625

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☐	Status	Name ↑	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☐	✔️	managementnet-vm-1	us-central1-c			10.130.0.2 (nic0)	34.28.248.151 (nic0)	SSH ▾ ⋮
☐	✔️	mynet-vm-1	us-central1-c			10.128.0.2 (nic0)	34.132.85.179 (nic0)	SSH ▾ ⋮
☐	✔️	mynet-vm-2	europa-west1-b			10.132.0.2 (nic0)	34.78.241.167 (nic0)	SSH ▾ ⋮
☐	✔️	privatenet-vm-1	us-central1-c			172.16.0.2 (nic0)	34.59.31.233 (nic0)	SSH ▾ ⋮

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What do you want to learn today?

Multiple VPC Networks

Lab setup instructions and requirements

End Lab00:39:52

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Usernamestudent-01-efbc7dd07c214

PasswordH0jjum9t8pUN

Project IDqwiklabs-gcp-04-a50ab16:

3. For mynet-vm-1, click SSH to launch a terminal and connect.

4. To test connectivity to mynet-vm-2's external IP, run the following command, replacing mynet-vm-2's external IP:

ping -c 3 'Enter mynet-vm-2 external IP here'

This should work!

5. To test connectivity to managementnet-vm-1's external IP, run the following command, replacing managementnet-vm-1's external IP:

ping -c 3 'Enter managementnet-vm-1 external IP here'

This should work!

6. To test connectivity to privatenet-vm-1's external IP, run the following command, replacing privatenet-vm-1's external IP:

Objectives

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Multiple VPC Networks

Lab setup instructions and requirements

End Lab00:38:32

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Open Google Cloud console

Username

student-01-efbc7dd07c214

Password

H0jjum9t8pUN

Project ID

qwiklabs-gcp-04-a50ab16

Multiple VPC Networks

ping -c 3 'Enter mynet-vm-2 external IP here'

This should work!

5. To test connectivity to managementnet-vm-1's external IP, run the following command, replacing managementnet-vm-1's external IP:

ping -c 3 'Enter managementnet-vm-1 external IP here'

This should work!

6. To test connectivity to privatenet-vm-1's external IP, run the following command, replacing privatenet-vm-1's external IP:

ping -c 3 'Enter privatenet-vm-1 external IP here'

This should work!

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EXTERNAL:

mynet-vm-2, 34.78.241.167

managementnet-vm-1, 34.28.248.151

and privatenet-vm-1 34.59.31.233

INTERNAL:

mynet-vm-2, 10.132.0.2

managementnet-vm-1, 10.130.0.2

and privatenet-vm-1 172.16.0.2

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End Lab 00:33:57

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Open Google Cloud console

Username student-01-efbc7dd07c214

Password H0jjum9t8pUN

Project ID qwiklabs-gcp-04-a50ab16:

Ping the internal IP addresses

Ping the internal IP addresses of the VM instances to determine if you can reach the instances from within a VPC network.

1. In the Cloud console, navigate to **Navigation menu > Compute Engine > VM instances**.

2. Note the internal IP addresses for **mynet-vm-2**, **managementnet-vm-1**, and **privatenet-vm-1**.

3. Return to the **SSH** terminal for **mynet-vm-1**.

4. To test connectivity to **mynet-vm-2**'s internal IP, run the following command, replacing **mynet-vm-2**'s internal IP:

ping -c 3 'Enter mynet-vm-2 internal IP here'

Note: You are able to ping the internal IP address of **mynet-vm-2** because it is on

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Multiple VPC Networks

privatenet-vm-1.

3. Return to the SSH terminal for mynet-vm-1.

4. To test connectivity to mynet-vm-2's internal IP, run the following command, replacing mynet-vm-2's internal IP:

ping -c 3 'Enter mynet-vm-2 internal IP here'

Note: You are able to ping the internal IP address of mynet-vm-2 because it is on the same VPC network as the source of the ping (mynet-vm-1), even though both VM instances are in separate zones, regions and continents!

5. To test connectivity to managementnet-vm-1's internal IP, run the following command, replacing managementnet-vm-1's internal IP:

ping -c 3 'Enter managementnet-vm-1 internal IP here'

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End Lab00:31:34

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Usernamestudent-01-efbc7dd07c21d

PasswordH0jjum9t8pUN

Project IDqwiklabs-gcp-04-a50ab16d

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Lab setup instructions and requirements

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Open Google Cloud console

Username

student-01-efbc7dd07c214

Password

H0jjum9t8pUN

Project ID

qwiklabs-gcp-04-a50ab16

5. To test connectivity to **managementnet-vm-1**'s internal IP, run the following command, replacing **managementnet-vm-1**'s internal IP:

ping -c 3 'Enter managementnet-vm-1 internal IP here'

Note: This should not work as indicated by a 100% packet loss!

6. To test connectivity to **privatenet-vm-1**'s internal IP, run the following command, replacing **privatenet-vm-1**'s internal IP:

ping -c 3 'Enter privatenet-vm-1 internal IP here'

Note: This should not work either as indicated by a 100% packet loss! You are unable to ping the internal IP address of **managementnet-vm-1** and **privatenet-vm-1** because they are in separate VPC networks from the source of the ping (**mynet-vm-1**), even though they are all in the same region **us-central1**.

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```
3 packets transmitted, 0 received, 100% packet loss, time 2034ms

student-01-efbc7dd07c21@mynnet-vm-1:~$
```

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Open Google Cloud console

Usernamestudent-01-efbc7dd07c214

PasswordH0jjum9t8pUN

Project IDqwiklabs-gcp-04-a50ab16:

Task 4. Create a VM instance with multiple network interfaces

Every instance in a VPC network has a default network interface. You can create additional network interfaces attached to your VMs. Multiple network interfaces enable you to create configurations in which an instance connects directly to several VPC networks (up to 8 interfaces, depending on the instance's type).

Create the VM instance with multiple network interfaces

Create the **vm-appliance** instance with network interfaces in **privatesubnet-1**, **managementsubnet-1** and **mynetwork**. The CIDR ranges of these subnets do not overlap, which is a requirement for creating a VM with multiple network interface controllers (NICs).

1. In the Cloud console, navigate to **Navigation menu > Compute Engine > VM**

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controllers (NICs).

Lab setup instructions and requirements

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Open Google Cloud console

Username

student-01-efbc7dd07c214

Password

H0jjum9t8pUN

Project ID

qwiklabs-gcp-04-a50ab16

1. In the Cloud console, navigate to **Navigation menu > Compute Engine > VM instances**.

2. Click **Create Instance**.

3. In the **Machine configuration**:

Set the following values, leave all other values at their defaults:

Property	Value (type value or select option as specified)
Name	vm-appliance
Region	us-central1
Zone	us-central1-c
Series	E2
Machine Type	e2-standard-4

Note: The number of interfaces allowed in an instance is dependent on the

Objectives

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Firewall policies – Network Security –

Compute Engine – qwiklabs-gcp x

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• Machine configuration
e2-standard-4, us-central1-c

• OS and storage
Debian GNU/Linux 12 (bookworm)

• Data protection
Snapshot schedules

• Networking
2 network interfaces, privatesubnet-1 (172.16.0.0/24)

• Observability
Install Ops Agent

• Security

• Advanced

Machine configuration

Name *
vm-appliance ?

Region *
us-central1 (Iowa) ?
Region is permanent

Zone *
us-central1-c ?
Zone is permanent

✓ General purpose

Compute optimized

Memory optimized

Storage optimized

GPUs

Machine types for common workloads, optimized for cost and flexibility

	Series ?	Description	vCPUs ?	Memory ?	CPU Platform
☐	C4	Consistently high performance	2 - 288	4 - 2,232 GB	Intel Emerald
☐	C4A	Arm-based consistently high performance	1 - 72	2 - 576 GB	Google Axion
☐	C4D	Consistently high performance	2 - 384	3 - 3,072 GB	AMD Turin
☐	N4	Flexible & cost-optimized	2 - 80	4 - 640 GB	Intel Emerald
☐	N4D	Flexible & cost-optimized	2 - 96	4 - 768 GB	AMD Turin
☐	C3	Consistently high performance	4 - 192	8 - 1,536 GB	Intel Sapphire
☐	C3D	Consistently high performance	4 - 360	8 - 2,880 GB	AMD Genoa
☒	E2	Low cost, day-to-day computing	0.25 - 32	1 - 128 GB	Intel Broadwe
☐	N2	Balanced price & performance	2 - 128	2 - 864 GB	Intel Cascade
☐	N2D	Balanced price & performance	2 - 224	2 - 896 GB	AMD Milan
☐	T2A	Scale-out workloads	1 - 48	4 - 192 GB	Ampere Altra

Create

Cancel

🔗 Equivalent code

Monthly estimate

\$98.84

That's about \$0.14 hourly

Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
4 vCPU + 16 GB memory	\$97.84
10 GB balanced persistent disk	\$1.00
Logging	Cost varies 🔗
Monitoring	Cost varies 🔗
Snapshot schedule	Cost varies 🔗
Total	\$98.84

[Compute Engine pricing](#) 🔗

[Cloud Operations pricing](#) 🔗

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• Machine configuration

e2-standard-4, us-central1-c

• OS and storage

Debian GNU/Linux 12 (bookworm)

• Data protection

Snapshot schedules

• Networking

2 network interfaces, privatesubnet-1 (172.16.0.0/24)

• Observability

Install Ops Agent

• Security

• Advanced

N4

Flexible & cost-optimized

2 - 80

4 - 640 GB

Intel Emerald

N4D

Flexible & cost-optimized

2 - 96

4 - 768 GB

AMD Turin

C3

Consistently high performance

4 - 192

8 - 1,536 GB

Intel Sapphire

C3D

Consistently high performance

4 - 360

8 - 2,880 GB

AMD Genoa

E2

Low cost, day-to-day computing

0.25 - 32

1 - 128 GB

Intel Broadwe

N2

Balanced price & performance

2 - 128

2 - 864 GB

Intel Cascade

N2D

Balanced price & performance

2 - 224

2 - 896 GB

AMD Milan

T2A

Scale-out workloads

1 - 48

4 - 192 GB

Ampere Altra

T2D

Scale-out workloads

1 - 60

4 - 240 GB

AMD Milan

N1

Balanced price & performance

0.25 - 96

0.6 - 624 GB

Intel Haswell

Machine type

Choose a machine type with preset amounts of vCPUs and memory that suit most workloads. Or, you can create a custom machine for your workload's particular needs. [Learn more](#)

Preset

Custom

e2-standard-4 (4 vCPU, 2 core, 16 GB memory)

▼

vCPU

4 (2 cores)

Memory

16 GB

✓ Advanced configurations

Monthly estimate

\$98.84

That's about \$0.14 hourly

Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
4 vCPU + 16 GB memory	\$97.84
10 GB balanced persistent disk	\$1.00
Logging	Cost varies
Monitoring	Cost varies
Snapshot schedule	Cost varies
Total	\$98.84

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☐	✓	mynet-vm-1	us-central1-c			10.128.0.2 (nic0)	34.132.85.179 (nic0)	SSH ▾ ⋮
☐	✓	mynet-vm-2	europa-west1-b			10.132.0.2 (nic0)	34.78.241.167 (nic0)	SSH ▾ ⋮
☐	✓	privatenet-vm-1	us-central1-c			172.16.0.2 (nic0)	34.59.31.233 (nic0)	SSH ▾ ⋮
☐	✓	vm-appliance	us-central1-c			172.16.0.4 (nic0) 10.130.0.4 (nic1) 10.128.0.3 (nic2)	34.61.137.200 (nic0) 35.232.56.73 (nic1) 34.9.105.110 (nic2)	SSH ▾ ⋮

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Lab setup instructions and requirements

End Lab00:17:37

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[Learn more.](#)

Open Google Cloud console

Username

student-01-efbc7dd07c214

Password

H0jjum9t8pUN

Project ID

qwiklabs-gcp-04-a50ab161

within that subnet (10.128.0.0/20), and has applicable firewall rules.

Note: Each network interface has its own internal IP address so that the VM instance can communicate with those networks.

8. In the Cloud console, navigate to **Navigation menu > Compute Engine > VM instances**.

9. For **vm-appliance**, click **SSH** to launch a terminal and connect.

10. Run the following, to list the network interfaces within the VM instance:

sudo ifconfig

Copied!

The output should look like this:

```
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1460
      inet 172.16.0.3 netmask 255.255.255.255 broadcast 172.16.0.3
      inet6 fe80::4281:c0ff:fe18:3 prefixlen 64 scopeid 0x20<link>
```

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Task 4. Create a VM instance with multiple network interfaces

Congratulations!

100/100


```
ping -c 3 172.16.0.2
```

```
ping -c 3 172.16.0.2
```

```
ping -c 3 10.130.0.2
```

```
ping -c 3 10.128.0.2
```

```
ping -c 3 10.132.0.2
```

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What do you want to learn today?

Multiple VPC Networks

Explore the network interface connectivity

Demonstrate that the **vm-appliance** instance is connected to **privatesubnet-1**, **managementsubnet-1** and **mynetwork** by pinging VM instances on those subnets.

1. In the Cloud console, navigate to **Navigation menu > Compute Engine > VM instances**.

2. Note the internal IP addresses for **privatenet-vm-1**, **managementnet-vm-1**, **mynet-vm-1**, and **mynet-vm-2**.

3. Return to the **SSH** terminal for **vm-appliance**.

4. To test connectivity to **privatenet-vm-1**'s internal IP, run the following command, replacing **privatenet-vm-1**'s internal IP:

ping -c 3 'Enter privatenet-vm-1's internal IP here'

This works!

Lab setup instructions and requirements

End Lab

00:12:56

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Username

student-01-efbc7dd07c214

Password

H0jjum9t8pUN

Project ID

qwiklabs-gcp-04-a50ab16

Objectives

Setup and requirements

Task 1. Create custom mode VPC networks with firewall rules

Task 2. Create VM instances

Task 3. Explore the connectivity between VM instances

Task 4. Create a VM instance with multiple network interfaces

Congratulations!

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What do you want to learn today?

Multiple VPC Networks

replacing **privatenet-vm-1**'s internal IP:

Lab setup instructions and requirements

End Lab00:12:41

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Username

student-01-efbc7dd07c214

Password

H0jjum9t8pUN

Project ID

qwiklabs-gcp-04-a50ab16:

ping -c 3 'Enter privatenet-vm-1's internal IP here'

This works!

5. Repeat the same test by running the following:

ping -c 3 privatenet-vm-1

Note: You are able to ping **privatenet-vm-1** by its name because VPC networks have an internal DNS service that allows you to address instances by their DNS names rather than their internal IP addresses. When an internal DNS query is made with the instance hostname, it resolves to the primary interface (nic0) of the instance. Therefore, this only works for **privatenet-vm-1** in this case.

6. To test connectivity to **managementnet-vm-1**'s internal IP, run the following command, replacing **managementnet-vm-1**'s internal IP:

100/100

Objectives

Setup and requirements

Task 1. Create custom mode VPC networks with firewall rules

Task 2. Create VM instances

Task 3. Explore the connectivity between VM instances

Task 4. Create a VM instance with multiple network interfaces

Congratulations!

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What do you want to learn today?

Multiple VPC Networks

Lab setup instructions and requirements

End Lab00:12:27

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Username

student-01-efbc7dd07c214

Password

H0jjum9t8pUN

Project ID

qwiklabs-gcp-04-a50ab16:

Multiple VPC Networks

```
ping -c 3 'Enter mynet-vm-1's internal IP here'
```

This works!

8. To test connectivity to **mynet-vm-2**'s internal IP, run the following command, replacing **mynet-vm-2**'s internal IP:

```
ping -c 3 'Enter mynet-vm-2's internal IP here'
```

Note: This does not work! In a multiple interface instance, every interface gets a route for the subnet that it is in. In addition, the instance gets a single default route that is associated with the primary interface eth0. Unless manually configured otherwise, any traffic leaving an instance for any destination other than a directly connected subnet will leave the instance via the default route on eth0.

9. To list the routes for **vm-appliance** instance, run the following command:

Objectives

Setup and requirements

Task 1. Create custom mode VPC networks with firewall rules

Task 2. Create VM instances

Task 3. Explore the connectivity between VM instances

Task 4. Create a VM instance with multiple network interfaces

Congratulations!

100/100

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```
student-01-efbc7dd07c21@vm-appliance:~$ ping -c 3 172.16.0.2
ping -c 3 172.16.0.2
ping -c 3 10.130.0.2
ping -c 3 10.128.0.2
ping -c 3 10.132.0.2
PING 172.16.0.2 (172.16.0.2) 56(84) bytes of data.
64 bytes from 172.16.0.2: icmp_seq=1 ttl=64 time=1.66 ms
64 bytes from 172.16.0.2: icmp_seq=2 ttl=64 time=0.364 ms
64 bytes from 172.16.0.2: icmp_seq=3 ttl=64 time=0.267 ms

--- 172.16.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2010ms
rtt min/avg/max/mdev = 0.267/0.763/1.658/0.634 ms
PING 172.16.0.2 (172.16.0.2) 56(84) bytes of data.
64 bytes from 172.16.0.2: icmp_seq=1 ttl=64 time=0.177 ms
64 bytes from 172.16.0.2: icmp_seq=2 ttl=64 time=0.335 ms
64 bytes from 172.16.0.2: icmp_seq=3 ttl=64 time=0.225 ms

--- 172.16.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2044ms
rtt min/avg/max/mdev = 0.177/0.245/0.335/0.066 ms
PING 10.130.0.2 (10.130.0.2) 56(84) bytes of data.
64 bytes from 10.130.0.2: icmp_seq=1 ttl=64 time=1.52 ms
64 bytes from 10.130.0.2: icmp_seq=2 ttl=64 time=0.287 ms
64 bytes from 10.130.0.2: icmp_seq=3 ttl=64 time=0.308 ms

--- 10.130.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2012ms
rtt min/avg/max/mdev = 0.287/0.704/1.517/0.574 ms
PING 10.128.0.2 (10.128.0.2) 56(84) bytes of data.
64 bytes from 10.128.0.2: icmp_seq=1 ttl=64 time=2.04 ms
64 bytes from 10.128.0.2: icmp_seq=2 ttl=64 time=0.391 ms
64 bytes from 10.128.0.2: icmp_seq=3 ttl=64 time=0.308 ms

--- 10.128.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2011ms
rtt min/avg/max/mdev = 0.308/0.912/2.037/0.796 ms
PING 10.132.0.2 (10.132.0.2) 56(84) bytes of data.

--- 10.132.0.2 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2044ms
```

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Multiple VPC Networks

Lab setup instructions and requirements

End Lab 00:12:04

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

Open Google Cloud console

Username

student-01-efbc7dd07c210

Password

H0jjum9t8pUN

Project ID

qwiklabs-gcp-04-a50ab161

route for the subnet that it is in. In addition, the instance gets a single default route that is associated with the primary interface eth0. Unless manually configured otherwise, any traffic leaving an instance for any destination other than a directly connected subnet will leave the instance via the default route on eth0.

9. To list the routes for **vm-appliance** instance, run the following command:

ip route

The output should look like this:

```
default via 172.16.0.1 dev eth0
10.128.0.0/20 via 10.128.0.1 dev eth2
10.128.0.1 dev eth2 scope link
10.130.0.0/20 via 10.130.0.1 dev eth1
10.130.0.1 dev eth1 scope link
172.16.0.0/24 via 172.16.0.1 dev eth0
172.16.0.1 dev eth0 scope link
```

Objectives

Setup and requirements

Task 1. Create custom mode VPC networks with firewall rules

Task 2. Create VM instances

Task 3. Explore the connectivity between VM instances

Task 4. Create a VM instance with multiple network interfaces

Congratulations!

Firewall policies – Network Security

VM instances – Compute Engine

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https://console.cloud.google.com/compute/instances?onCreate=true&project=qwiklabs-gcp-04-a50ab163c625

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VM instances

Create instance

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VM instances

Filter Enter property name or value

☐	Status	Name ↑
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☐	✓	mynet-vm-1
☐	✓	mynet-vm-2
☐	✓	privatenet-vm-1
☐	✓	vm-appliance

Related actions

ssh.cloud.google.com/v2/ssh/projects/qwiklabs-gcp-04-a50ab163c625/zones/us-central1-c/i/...

https://ssh.cloud.google.com/v2/ssh/projects/qwiklabs-gcp-04-a50ab163c625/zones/...

SSH-in-browser

📄 UPLOAD FILE

📄 DOWNLOAD FILE

🗨️ 📄 ⚙️

```
student-01-efbc7dd07c21@vm-appliance:~$ ip route
default via 172.16.0.1 dev ens4 proto dhcp src 172.16.0.4 metric 100
10.128.0.0/20 via 10.128.0.1 dev ens6 proto dhcp src 10.128.0.3 metric 100
10.128.0.1 dev ens6 proto dhcp scope link src 10.128.0.3 metric 100
10.130.0.0/20 via 10.130.0.1 dev ens5 proto dhcp src 10.130.0.4 metric 100
10.130.0.1 dev ens5 proto dhcp scope link src 10.130.0.4 metric 100
169.254.169.254 via 172.16.0.1 dev ens4 proto dhcp src 172.16.0.4 metric 100
172.16.0.0/24 via 172.16.0.1 dev ens4 proto dhcp src 172.16.0.4 metric 100
172.16.0.1 dev ens4 proto dhcp scope link src 172.16.0.4 metric 100
student-01-efbc7dd07c21@vm-appliance:~$
```

Connect

0)	SSH	▼	⋮
0)	SSH	▼	⋮
0)	SSH	▼	⋮
0)	SSH	▼	⋮
0)	35.232.56.73	SSH	▼
5.110	(nic2)		

Show

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